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Defined, *design* is “the arrangement of elements or details in a product or work of art.” Of course, given a large number of elements, they can be arranged in an almost endless way with a plethora of results. Many things can impact the design of your real-time character, but the most impacting and unavoidable one is *limitations*.

Limitations

By limitations I don’t mean limitations on your imagination. I mean hard limitations like making a convincing human or creature out of a thousand triangles or less. However, with the growth of the hardware and computing power of the PC and the next generation of consoles like Xbox 2 and PlayStation 3, polygon limitations are becoming less severe. Still, characters for real-time games can rarely afford to have flowing gowns or octopoid tentacles. Organic shapes that undulate and look “squishy” are hard to pull off, as is cloth or long, flowing strands of hair when you have to represent them with triangles. Designing real-time characters requires much more thought than a character destined *only* to be rendered, as in a movie. With lower-polygon meshes it’s usually best to keep the designs compact and simple using transparency mapped onto two-sided, segmented planes to create the illusion of hair or scraps of cloth.

Even as polygon limitations ease and characters that are 10,000 triangles or more are possible, other limitations like number of bones in a skeleton, number of possible morph targets, and amount of texture resolution will always exist. For example, this real-time character done for a recent XNA demo for the Xbox is over 80,000 triangles:

定义上，设计是“产品或艺术品中元素或细节的排列。”当然，由于元素众多，它们可以以几乎无穷无尽的方式排列，产生多种结果。许多因素会影响你的实时角色的设计，但最具影响力且不可避免的因素是局限。

局限

我说的局限不是指想象力的局限，而是硬性限制，比如用一千个甚至更少的三角形塑造一个令人信服的人类或生物。然而，随着个人电脑硬件和计算能力的增长，以及Xbox 2和PlayStation 3等下一代游戏主机的出现，多边形限制正变得越来越宽松。尽管如此，实时游戏的角色通常还是难以拥有飘逸的长袍或章鱼触手。那些起伏不定、看起来“软绵绵”的有机形态很难实现，布料或长而飘逸的头发也不例外，当你需要用三角形来表现它们时。设计实时角色需要比电影中只用于渲染的角色更多的思考。使用低多边形网格时，通常最好保持设计紧凑和简单，利用透明度映射到双面分割平面上，来营造头发或碎布片的错觉。

即使多边形限制逐渐放宽，角色可以达到10000个甚至更多三角形，其他限制如骨架中的骨骼数量、可能的形态目标数量以及纹理分辨率等始终存在。例如，这个为近期Xbox的XNA演示制作的实时角色拥有超过80000个三角形：